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Lab5

11/05/2025

Creating a VPC: Define an isolated network by creating a Virtual Private Cloud (VPC). This provides complete control over routing, subnets, and IP addresses. Every resource in this lab has private IP space reserved by the CIDR block 10.0.0.0/16.

The screenshot shows the AWS Management Console interface for a newly created VPC. At the top, a green notification bar states: "You successfully created vpc-012b5131313918a22 / lab5-vpc". The breadcrumb navigation shows: VPC > Your VPCs > vpc-012b5131313918a22. The main heading is "vpc-012b5131313918a22 / lab5-vpc" with an "Actions" button. Below this is a "Details" section with a grid of configuration items:

Details	
VPC ID vpc-012b5131313918a22	State Available
DNS resolution Enabled	Tenancy default
Main network ACL acl-033411cc25609edfa	Default VPC No
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled
Block Public Access Off	DHCP option set dopt-068fb8db19fdd941f
IPv4 CIDR 10.0.0.0/16	Route 53 Resolver DNS Firewall rule groups -
DNS hostnames Disabled	Main route table rtb-0d215587128124389
Owner ID 143790590778	

Below the details is a "Resource map" section with tabs for "Resource map", "CIDRs", "Flow logs", "Tags", and "Integrations". The "Resource map" tab is active, showing a summary of the VPC resources:

- VPC**: Your AWS virtual network
- Subnets (0)**: Subnets within this VPC
- Route tables (1)**: Route network traffic to resources

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Creating an internet gateway: The VPC is connected to the public internet via an Internet Gateway (IGW). Once routing and security rules are set up, attaching it to the VPC enables resources like EC2 instances to transmit and receive traffic outside of AWS.

The screenshot displays the AWS Management Console interface for an Internet Gateway. At the top, a green notification bar states: "Internet gateway igw-097fdeffd2469a90d successfully attached to vpc-012b5131313918a22". Below this, the page title is "igw-097fdeffd2469a90d / lab5-igw".

The **Details** section shows the following information:

- Internet gateway ID:** igw-097fdeffd2469a90d
- State:** Attached (indicated by a green checkmark)
- VPC ID:** vpc-012b5131313918a22 | lab5-vpc
- Owner:** 143790590778

The **Tags (1)** section shows a single tag:

Key	Value
Name	lab5-igw

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Creating Subnets: In separate Availability Zones (us-east-1a and us-east-1b), two public subnets are established. This increases fault tolerance and makes it possible for the load balancer to divide traffic among zones for high availability.

The screenshot shows the AWS Management Console interface for creating a new subnet. The breadcrumb navigation at the top indicates the path: VPC > Subnets > Create subnet. The account ID is 1437-9059... and the time is 11:05 AM.

Subnet 1 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
lab5-public-subnet-1a
The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / use1-az6 (us-east-1a)

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

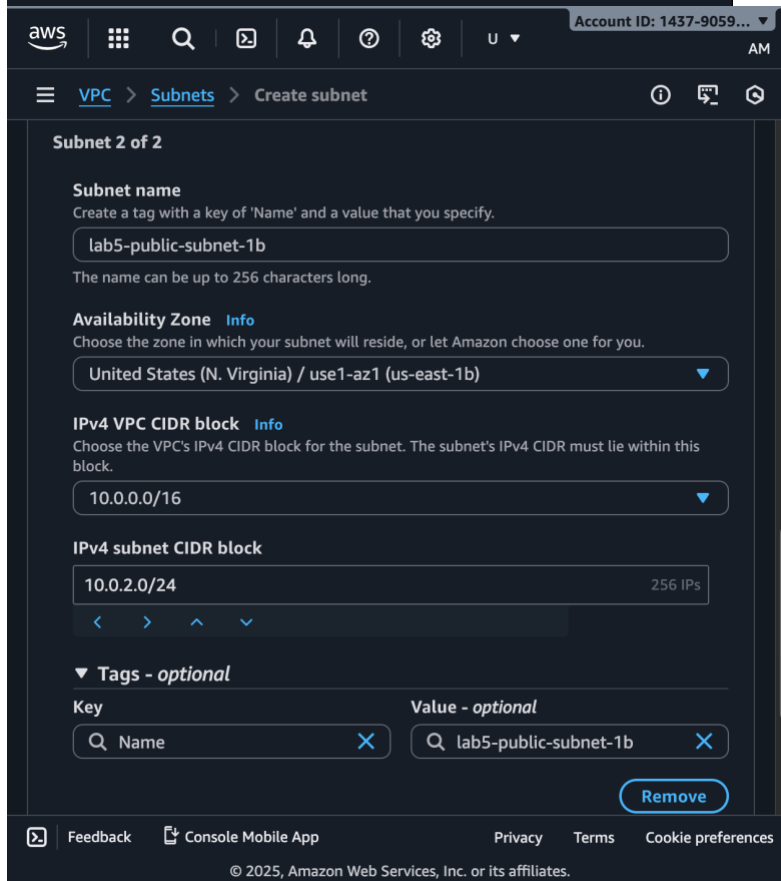
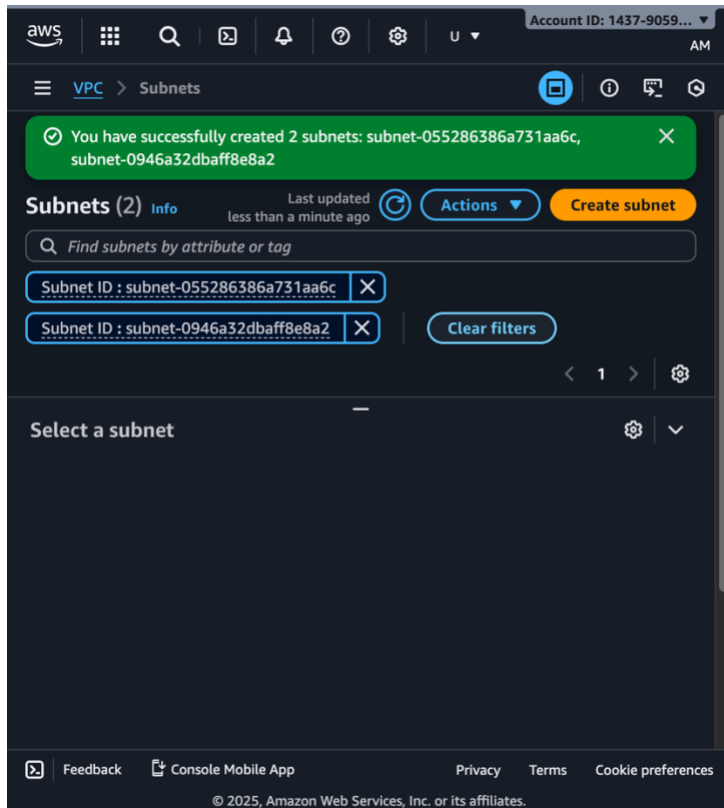
IPv4 subnet CIDR block
10.0.1.0/24 (256 IPs)
Navigation arrows: < > ^ v

Tags - optional

Key	Value - optional
Name	lab5-public-subnet-1a

[Remove](#)

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Creating a Route table: Network traffic is directed by the route table. Instances in these subnets will be able to access the internet if both subnets are associated and a route 0.0.0.0/0 pointing to the IGW is added.

The screenshot shows the AWS Management Console interface for the 'Route tables' section. The left sidebar contains navigation links for 'VPC dashboard', 'AWS Global View', and 'Virtual private cloud'. The main content area shows a list of route tables with columns for Name, Route table ID, Explicit subnet associations, Edge associations, and Main. The 'lab5-public-RT' route table is selected, and its details are displayed below the list. The details section includes the Route table ID, VPC, Owner ID, Main status, Explicit subnet associations, and Edge associations.

Name	Route table ID	Explicit subnet associations	Edge associations	Main
lab5-public-RT	rtb-012be313499b3e0da	2 subnets	-	No
-	rtb-0d215587128124389	-	-	Yes

rtb-012be313499b3e0da / lab5-public-RT

Details

Route table ID rtb-012be313499b3e0da	Main No	Explicit subnet associations 2 subnets	Edge associations -
VPC vpc-012b5131313918a22 lab5-vpc	Owner ID 143790590778		

Creating an EC2 Instance: A basic web server is an Ubuntu EC2 instance. Apache 2 installs automatically using user data and displays a simple HTML page with the hostname and IP. Afterwards, the load balancer uses this instance as a backend target.

The screenshot displays the AWS Management Console interface for the EC2 Instances page. At the top, the navigation bar includes the AWS logo, a search icon, a list icon, a notification bell, a help icon, a settings gear, and a user profile icon labeled 'U'. The account ID '1437-9059...' is visible in the top right corner. Below the navigation bar, the breadcrumb 'EC2 > Instances' is shown. The main content area is titled 'Instances (1/1) Info'. It features several action buttons: a refresh icon, 'Connect', 'Instance state' (with a dropdown arrow), 'Actions' (with a dropdown arrow), and a prominent orange 'Launch instances' button. A search bar prompts 'Find Instance by attribute or tag (case-sensitive)', and a filter dropdown is set to 'All states'. A table lists the instance 'lab-ec2' with ID 'i-0191c098b2c313e40' and state 'Running'. Below the table, the details for 'i-0191c098b2c313e40 (lab-ec2)' are expanded, showing the 'Instance summary' with fields for Instance ID, Public IPv4 address (35.153.127.36), Private IPv4 addresses (10.0.1.150), Instance state (Running), IPv6 address, and Public DNS.

aws Account ID: 1437-9059... AM

EC2 > Instances

Instances (1/1) Info

Refresh Connect Instance state Actions

Launch instances

Find Instance by attribute or tag (case-sensitive) All states

< 1 > ⚙️

☒	Name	Instance ID	Instance state	In:
☒	lab-ec2	i-0191c098b2c313e40	Running	t3.

i-0191c098b2c313e40 (lab-ec2) ⚙️

▼ Instance summary Info

Instance ID i-0191c098b2c313e40	Public IPv4 address 35.153.127.36 open address
Private IPv4 addresses 10.0.1.150	IPv6 address -
Instance state Running	Public DNS -

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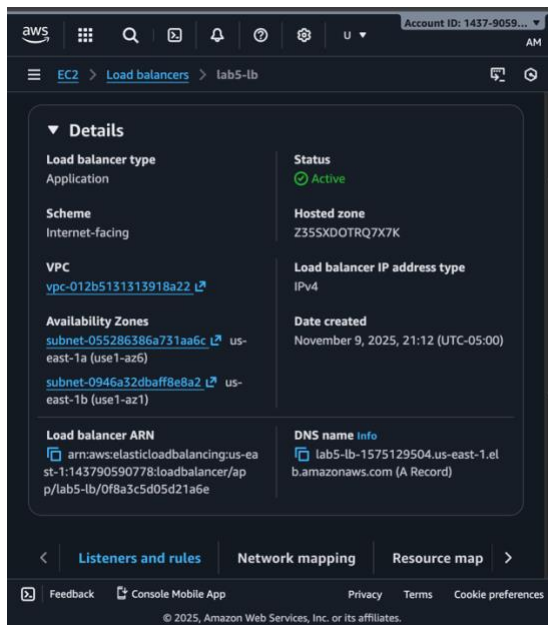
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Creating a Target Group: EC2 instances are logically grouped for the load balancer by a target group. It routes HTTP traffic to them and keeps an eye on their health. The target in this case is the single instance lab5-ec2.

The screenshot displays the AWS Management Console interface for a Target Group. At the top, a green notification banner states: "Successfully created the target group: lab5-ec2-tg. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab." The breadcrumb navigation shows the path: EC2 > Target groups > lab5-ec2-tg. The main heading is "lab5-ec2-tg" with an "Actions" dropdown menu. Below this, the "Details" section shows the ARN: `arn:aws:elasticloadbalancing:us-east-1:143790590778:targetgroup/lab5-ec2-tg/0c076...`. The details are organized into two columns: the left column lists "Target type" (Instance), "Protocol version" (HTTP1), and "IP address type" (IPv4); the right column lists "Protocol : Port" (HTTP: 80), "VPC" ([vpc-012b5131313918a22](#)), and "Load balancer" (None associated). At the bottom, a summary row shows: 1 Total targets, 0 Healthy (with a green checkmark icon), 0 Unhealthy (with a red X icon), and 1 Unused (with a grey circle icon). Below the healthy count, it also shows 0 Anomalous. The footer contains links for Feedback, Console Mobile App, Privacy, Terms, and Cookie preferences, along with the copyright notice: © 2025, Amazon Web Services, Inc. or its affiliates.

lab5-ec2-tg			
Details arn:aws:elasticloadbalancing:us-east-1:143790590778:targetgroup/lab5-ec2-tg/0c076...			
Target type Instance	Protocol : Port HTTP: 80		
Protocol version HTTP1	VPC vpc-012b5131313918a22		
IP address type IPv4	Load balancer None associated		
1 Total targets	0 Healthy	0 Unhealthy	1 Unused
	0 Anomalous		

Creating a Load Balancer: The Application Load Balancer (ALB) distributes incoming HTTP requests across targets in multiple subnets. It increases availability and provides a single DNS endpoint for clients to reach the backend servers securely.



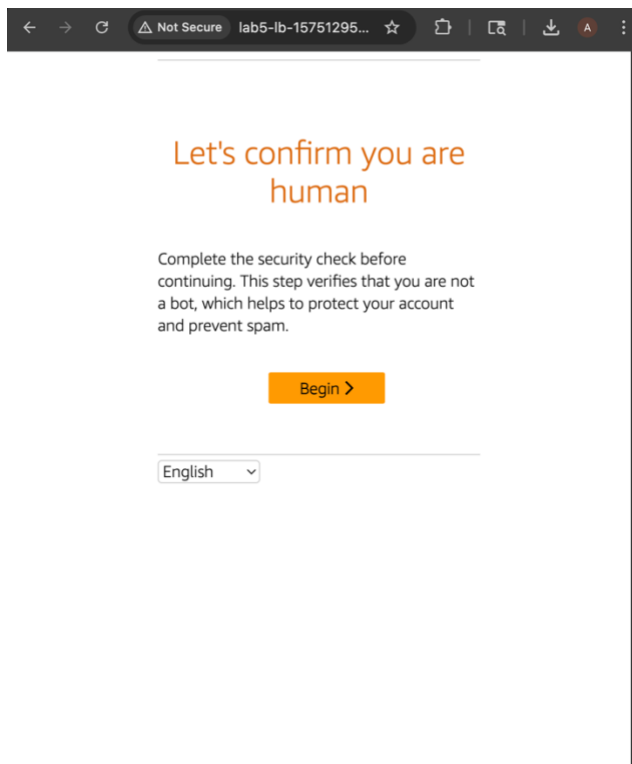
Creating a WAF: AWS WAF filters and manages HTTP requests to provide an additional degree of security. In this lab, a Web ACL with a rule preventing traffic from a particular IP set is made and connected to the ALB. This illustrates how WAF can employ rules to ban, permit, or challenge (CAPTCHA) users.

The screenshot displays the AWS WAF console interface. The left pane shows the 'Web ACLs' section with a table listing one Web ACL named 'lab5-aws-waf' in the 'US East (N. Virginia)' region. The right pane shows the details for the rule 'block-myip' associated with this Web ACL. The rule is a 'Regular rule' in the 'US East (N. Virginia)' region. Under the 'If a request matches the statement' section, 'Statement 1' is defined with the following details:

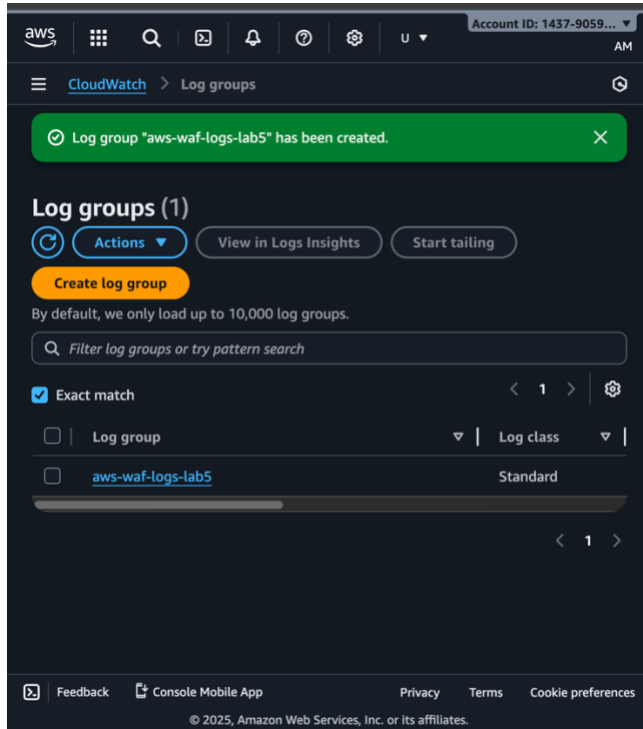
Field	Value
Rule name	block-myip
Type	Regular rule
Region	US East (N. Virginia)
Statement 1 IP set name	lab5-myip
Statement 1 IP set description	None
Statement 1 IP set ARN	arn:aws:wafv2:us-east-1:143790590778:regional/ipset/lab5-myip/7ba53d89-20d6-4a9c-bc81-0197712b0fab

Verifying the Working of WAF: The WAF configuration is tested by gaining access to the ALB's DNS name. WAF is actively blocking the specified IP, as seen by the "403 Forbidden" response. Modifying the rule action to Allow and CAPTCHA demonstrates dynamic control over the processing of requests.

403 Forbidden



Creating CloudWatch Logs: CloudWatch Logs collects and stores WAF activity for analysis. Creating a log group like aws-waf-logs-lab5 lets administrators monitor requests, troubleshoot security events, and visualize blocked or allowed traffic patterns.



Attaching and Viewing CloudWatch Logs to the WAF: Enabling logging on the WAF sends all web ACL events to CloudWatch. Viewing the log streams verifies that WAF actions are recorded, proving end-to-end monitoring and auditing are active.

